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BE (8th Semester)

Examination, April - May, 2017

[New Scheme]

Non-Conventional Energy Sources

Time Allowed : 3 hours *Maximum Marks :* 80

Minimum Pass Marks : 28

Note : (*i*) Attempt **All** questions. Part (*a*) of each question is compulsory. Attempt any **two** parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) What are primary and secondary energy sources? [2]

(*b*) Why orientation is needed in concentrating type collectors? Describe the different methods of Sun tracking. [7]

(*c*) What are the advantages and disadvantages of concentrating collectors over a flat-plate collectors? [7]

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(*d*) Classify different types of solar thermal collectors and show the constructional details of a flat-plate collector. [7]

2. (*a*) Write different main applications of solar energy. [2]

(*b*) Write short notes on : [7]

(*i*) Agriculture and industrial process heat

(*ii*) Solar pumping

(*c*) What do you mean by greenhouse? Enumerate the main types of greenhouses. [7]

(*d*) Describe a passive solar space heating system (any one type). [7]

3. (*a*) Explain the process of "photosynthesis". [2]

(*b*) Explain the constructional detail and working of KVIC digester. [7]

(*c*) Explain in detail biomass conversion technologies. [7]

(*d*) Write different usable forms of biomass, there compositions and fuel properties. [7]

4. (*a*) What is the basic principle of wind energy conversion? [2]

(*b*) Describe with a neat sketch the working of a wind energy conversion system (WECS) with main components. [7]

(*c*) Explain with sketches the various methods of tidal power generation. What are the limitations of each method? [7]

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(*d*) Write a short note on wave energy conversion machines. What are the advantages and limitations of wave energy conversion? [7]

5. (*a*) Write the main applications of hydrogen gas. [2]

(*b*) Explain various methods of production of hydrogen for use as energy carrier (any two methods). [7]

(*c*) Discuss in detail about design, classification and application of a fuel cell. [7]

(*d*) Explain various methods of storage and transportation of hydrogen. [7]

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